

Blood Related Disorders

What is haemophilia?

Haemophilia is an inherited bleeding disorder, which means it can be passed from parents to their children. If you have haemophilia, your blood doesn't clot properly. This can lead to bleeding that is difficult to control.

In your blood, there are special proteins called 'clotting factors'. When a blood vessel is injured, clotting factors control blood loss by plugging or patching up the injury. People with haemophilia have lower than normal levels of a clotting factor.

Haemophilia is rare. It usually affects males, and nearly all people with severe haemophilia are males.

What are the types of haemophilia?

There are 2 types of haemophilia:

- **Haemophilia A** (also called classic haemophilia) is the most common type — it is caused by a lack of clotting factor 8.
- **Haemophilia B** (sometimes called Christmas disease) is caused by a lack of clotting factor 9.

It's important to know the type of haemophilia you have, as each type requires different clotting factor treatment.

Both types of haemophilia show the same symptoms. Some people have mild haemophilia, while others are more severely affected.

What are the symptoms of haemophilia?

The main signs of haemophilia are:

- easy bruising from an early age
- internal bleeding for no obvious reason, especially in your joints and muscles
- greater than normal bleeding after injuries or surgery

Haemophilia is rare in females, and females who are affected usually have mild disease. But some females with haemophilia can have:

- heavy or long periods, which can cause iron deficiency or anaemia
- heavy bleeding after giving birth

What causes haemophilia?

Haemophilia is an inherited condition and usually occurs in families.

Haemophilia is caused by a mutation in a gene that is located on the X chromosome.

Females have 2 copies of the X chromosome in their cells. Males have one X chromosome and one Y chromosome. Males who inherit a gene for haemophilia will be affected by haemophilia.

Females who inherit a gene for haemophilia often have an unaffected gene on their other X chromosome. This is called being a genetic carrier. This means that they can pass on the haemophilia gene to their children, but they may not have haemophilia themselves. For this reason, haemophilia is much more common in males.

Sometimes, haemophilia can be acquired. This occurs when a person's immune system targets their own clotting factors. Acquired haemophilia is very rare and its cause is not known. It may be triggered by another medical condition. This type of haemophilia affects males and females equally.

Cardiac Procedures

What is angioplasty?

Angioplasty is a minimally invasive procedure that opens arteries (large blood vessels) to let blood flow more easily. It most commonly treats atherosclerosis (a collection of plaque made of fat and cholesterol) in your arteries.

Angioplasty involves a tiny medical balloon that pushes plaque through in spots that are too narrow or blocked.

Angioplasties are generally safe procedures. They're less invasive than other heart and vascular procedures, like bypass surgery.

Your cardiovascular health is very important to your overall health.

Angioplasties are often necessary to try to prevent serious complications of atherosclerosis, like heart attack and stroke

- **Types of angioplasty**

-

- **Coronary artery angioplasty.** The coronary arteries are major blood vessels that supply blood to your heart. Another name for this procedure is percutaneous coronary intervention (PCI).

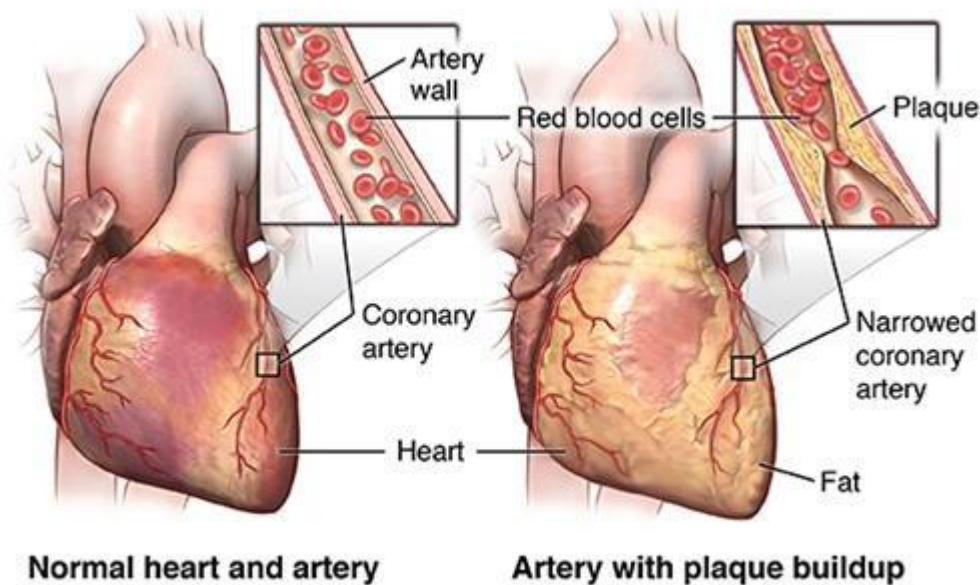
- **Peripheral artery angioplasty.** Peripheral arteries are blood vessels that carry blood away from your heart to your extremities, like your legs and arms.

Different angioplasty techniques include:

- **Balloon angioplasty.** Most angioplasty procedures involve a tiny medical balloon to move the plaque out of the way and off to the side of the artery wall.
- **Angioplasty with or without stenting.** Your provider may or may not place a permanent stent (a small tube) in the newly opened artery to help keep it open. Stent placement comes after most angioplasties.
- **Excimer laser coronary angioplasty (ELCA).** This is an alternative to using a balloon to unblock a coronary artery. The laser destroys the plaque. It's not as common as balloon angioplasty.

What is Coronary bypass graft surgery?

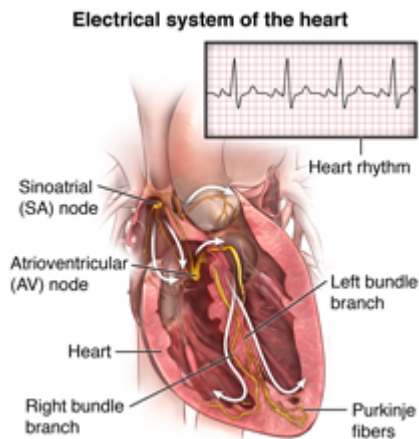
Coronary artery bypass graft surgery (CABG) is a procedure used to treat coronary artery disease. Coronary artery disease (CAD) is the narrowing of the coronary arteries. These are the blood vessels that supply oxygen and nutrients to the heart muscle. CAD is caused by a build-up of fatty material within the walls of the arteries. This buildup narrows the inside of the arteries, limiting the supply of oxygen-rich blood to the heart muscle.



One way to treat the blocked or narrowed arteries is to bypass the blocked portion of the coronary artery with a piece of a healthy blood vessel from elsewhere in your body. Blood vessels, or grafts, used for the bypass procedure may be pieces of a vein from your leg or an artery in your chest. An artery from your wrist may also be used. Your healthcare provider attaches 1 end of the graft above the blockage and the other end below the blockage. Blood bypasses the blockage by going through the new graft to reach the heart muscle. This is called coronary artery bypass surgery.

What is a pacemaker insertion?

A pacemaker insertion is the implantation of a small electronic device that is usually placed in the chest (just below the collarbone) to help regulate slow electrical problems with the heart. A pacemaker may be recommended to ensure that the heartbeat does not slow to a dangerously low rate.



The heart's electrical system

The heart is basically a pump made up of muscle tissue that is stimulated by electrical currents, which normally follow a specific circuit within the heart.

This normal electrical circuit begins in the sinus or sinoatrial (SA) node, which is a small mass of specialized tissue located in the right atrium (upper chamber) of the heart. The SA node generates an electrical stimulus at 60 to 100 times per minute (for adults) under normal conditions; this electrical impulse from the SA node starts the heartbeat.

The electrical impulse travels from the SA node via the atria to the atrioventricular (AV) node in the bottom of the right atrium. From there the impulse continues down an electrical conduction pathway called the Bundle of His and then on through the "His-Purkinje" system into the ventricles (lower chambers) of the heart. When the electrical stimulus occurs it causes the muscle to contract and pump blood to the rest of the body. This process of electrical stimulation followed by muscle contraction is what makes the heart beat.

Heart Transplant

A heart transplant is an operation in which a failing heart is replaced with a healthier donor heart. Heart transplant is a treatment that's usually reserved for people whose condition hasn't improved enough with medications or other surgeries.

Heart transplants are performed when other treatments for heart problems haven't worked, leading to heart failure. In adults, heart failure can be caused by:

- A weakening of the heart muscle (cardiomyopathy)
- Coronary artery disease
- Heart valve disease
- A heart problem you're born with (congenital heart defect)
- Dangerous recurring abnormal heart rhythms (ventricular arrhythmias) not controlled by other treatments
- Failure of a previous heart transplant

In children, heart failure is most often caused by either a congenital heart defect or cardiomyopathy.

Another organ transplant may be performed at the same time as a heart transplant (multiorgan transplant) in people with certain conditions at select medical centers.

Elementary Diseases of Digestive system

Functional Dyspepsia

Functional dyspepsia is a kind of chronic indigestion. It involves frequent stomachaches and a feeling of fullness or bloating after eating. A healthcare provider may diagnose you with functional dyspepsia if they can't find any direct cause for your symptoms. There are many treatment options, but there's no one solution that works for everyone.

What is functional dyspepsia?

Functional dyspepsia is chronic indigestion with no identifiable cause. "Dyspepsia" is another word for indigestion — a pattern of symptoms that occur together after eating and while you're digesting. "Functional" means that your dyspepsia symptoms can't be explained by the usual physical causes.

Typical symptoms of dyspepsia include burning stomach pain, over-fullness and bloating. These symptoms resemble peptic ulcer disease or GERD. But after testing, only a fraction of people with dyspepsia are diagnosed with these conditions. The rest are diagnosed with functional dyspepsia.

Types of functional dyspepsia

Dyspepsia is more of a syndrome than a condition — it's a pattern of symptoms that occur together. But different people can have different combinations of symptoms with it. Healthcare providers have identified two distinct ways that people can experience functional dyspepsia. They classify these as:

- **Epigastric pain syndrome.** This type is characterized by upper abdominal pain and burning, such as heartburn. It resembles symptoms of GERD and it doesn't necessarily occur at mealtimes.
- **Postprandial distress syndrome.** This type is characterized by early satiety and fullness, bloating and stomach pain after eating ("postprandial"). It resembles symptoms of peptic ulcer disease.

Hepatitis happens when something causes inflammation in your liver. You can spread hepatitis if you have close contact with someone who has the disease. Common symptoms include different types of upper abdominal pain, diarrhea and fatigue. Treatment includes lifestyle changes, medication and a liver transplant.

What is hepatitis?

Hepatitis is inflammation in your liver. Inflammation is your body's response to an infection or injury. Many things can injure your liver and trigger hepatitis. Certain viral infections cause hepatitis. But you also can develop it if you have diseases that affect your liver or you're exposed to toxins like chemicals, alcohol or drugs.

The condition may cause symptoms that happen suddenly and then go away within six months (acute hepatitis). But you may have hepatitis that doesn't go away and slowly gets worse. This is chronic hepatitis.

Types of hepatitis

There are several types of hepatitis. Each type can be acute or chronic. Some types are more common than others:

- **Viral hepatitis.** This is the most common type of hepatitis. There are several types of viral hepatitis. The most common ones in the U.S. are hepatitis A, hepatitis B and hepatitis C. Hepatitis D and hepatitis E aren't as common.

- **Toxic hepatitis**. This is hepatitis you can get from exposure to chemicals. Alcohol-induced hepatitis and drug-induced hepatitis are other types of toxic hepatitis. Toxic hepatitis can be acute or chronic.
- **Autoimmune hepatitis**. This chronic hepatitis is rare. It happens when your immune system attacks liver tissues by mistake.

Gastro Intestinal Procedures

The Whipple procedure is an operation to treat tumors and other conditions in the pancreas, small intestine and bile ducts. It involves removing the head of the pancreas, the first part of the small intestine, the gallbladder and the bile duct.

The Whipple procedure is also called a pancreaticoduodenectomy. It's often used to treat pancreatic cancer that hasn't spread beyond the pancreas.

The Whipple procedure is a complex operation. It can have serious risks. However, this surgery often saves lives.

Related procedures

There are other types of pancreas operations. Your surgeon may talk with you about these other procedures in certain situations. Seek a second opinion from a specialized surgeon if needed. Related procedures include:

- **Surgery for tumors or other conditions in the body and tail of the pancreas.** The body and tail of the pancreas make up the left side of the organ. Surgery to remove the left side of the pancreas is called a distal pancreatectomy. During this procedure, the surgeon also might need to remove the spleen.
- **Surgery to remove the whole pancreas.** This is called total pancreatectomy. After the pancreas is removed, you'll take medicine to replace the hormones and enzymes made by the pancreas for the rest of your life.

- **Surgery for tumors that affect nearby blood vessels.** When a tumor in the pancreas grows to involve nearby blood vessels, a more-complex procedure might be needed. The procedure might need to involve taking out and rebuilding parts of the blood vessels. Few medical centers in the United States have surgeons trained to do these blood vessel operations safely.

Nissen Fundoplication

A Nissen fundoplication is a surgery to treat gastroesophageal reflux disease (GERD). During the procedure, a surgeon wraps the top of your stomach around the bottom of your esophagus. This reinforces your lower esophageal sphincter to keep stomach acid from backing up into your esophagus.

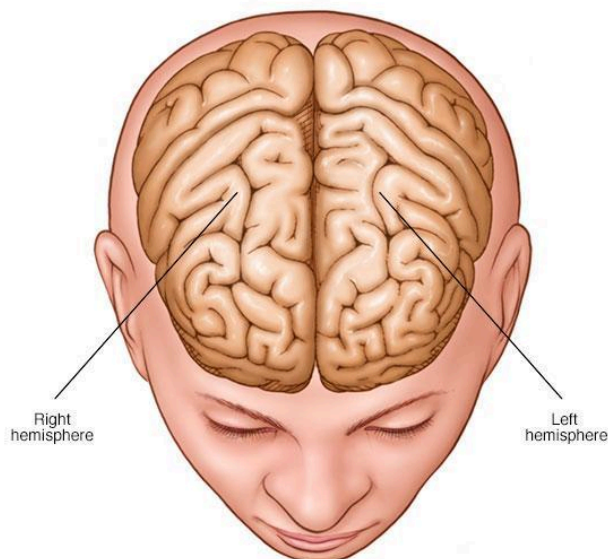
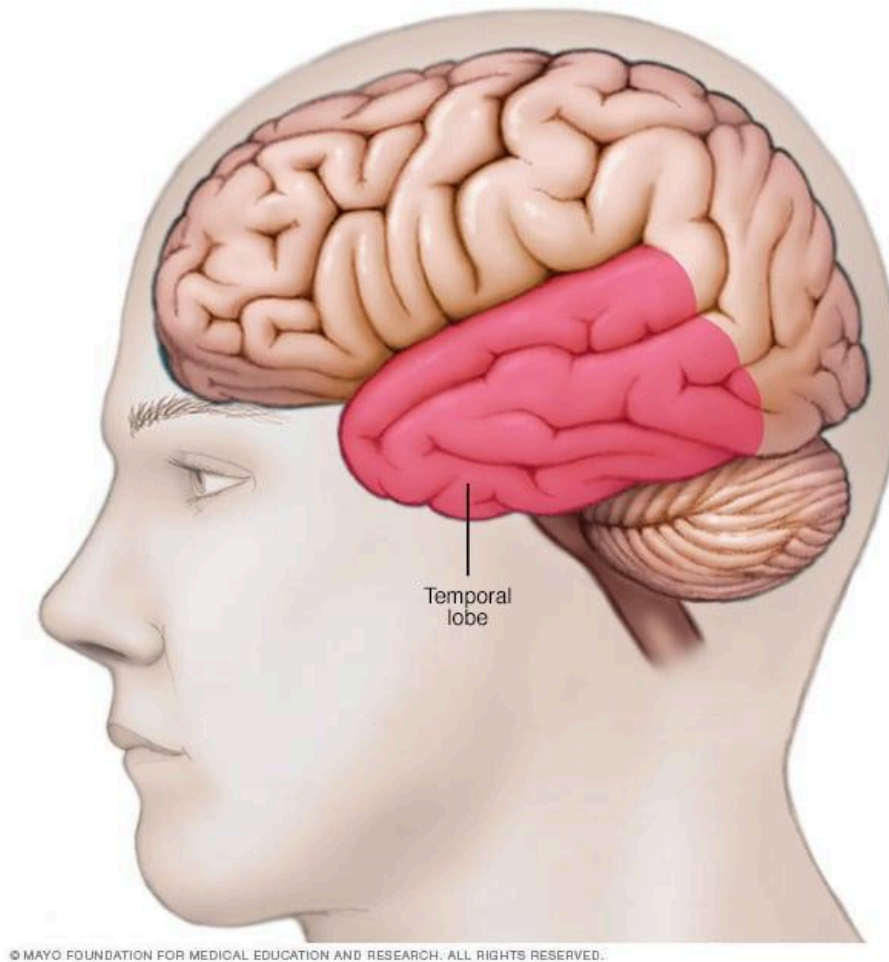
What is a Nissen fundoplication?

Nissen fundoplication is a surgery to treat gastroesophageal reflux disease (GERD). If you have GERD, the acid in your stomach flows back up to your esophagus. This can happen if your lower esophageal sphincter doesn't work as it should.

Normally, your sphincter relaxes to let food move from your esophagus to your stomach. It then tightens to keep stomach acid out. A Nissen procedure wraps the top of your stomach (fundus) around the bottom of your esophagus. The wrap reinforces your lower esophageal sphincter so stomach acid stays where it belongs.

NEURO PROCEDURES

Epilepsy Surgery



Epilepsy surgery may be an option when medicines aren't able to manage seizures. This condition is known as medically refractory epilepsy. It also is

called drug-resistant epilepsy. The goal of epilepsy surgery is to stop seizures or limit how bad they are.

After surgery, people generally need to remain on antiseizure medicines for at least two years. Over time, they may be able to lower the dose of their medicines or stop them completely.

Managing seizures is important because of the complications and health risks that can result if epilepsy isn't properly treated. Complications can include:

- Physical injuries during a seizure.
- Drowning, if the seizure occurs during a bath or swimming.
- Depression and anxiety.
- Developmental delays in children.
- Worsening memory or other thinking skills.
- Sudden death, a rare complication of epilepsy.

Types of epilepsy surgery

Epileptic seizures result from irregular activity of certain brain cells called neurons. The type of surgery to treat epileptic seizures depends on which area of the brain the seizures begin and the age of the person having the surgery. Types of surgery might include:

- **Resective surgery.** This surgery is the most common epilepsy surgery. With this surgery, the surgeon cuts out brain tissues in the area of the brain where seizures occur. The area may be the site of a tumor or brain injury. Resective surgery most often occurs on one of the temporal lobes. This area controls visual memory, language comprehension and emotions.
- **Laser interstitial thermal therapy (LITT).** This procedure is less invasive than resective surgery. LITT uses a laser to pinpoint and destroy a small portion of brain tissue. Magnetic resonance imaging (MRI) helps guide the surgeon's laser.
- **Deep brain stimulation.** This procedure involves implanting electrodes deep inside the brain. A pacemakerlike device also is placed under the skin in the chest. A wire connects this device to the electrodes in the brain. The electrodes send electrical impulses to disrupt seizure-causing activity. Like LITT, this procedure is guided by MRI.

- **Corpus callosotomy.** This surgery completely or partially removes part of the brain that connects nerves on the right and left sides of the brain. This area is called the corpus callosum. This procedure is performed most often on children who experience irregular brain activity that spreads from one side of the brain to the other.
- **Hemispherectomy.** This procedure removes one side of the brain, called a hemisphere, which is part of the folded gray matter of the brain. The gray matter area is called the cerebral cortex. This surgery is usually performed only in children who have seizures that start in different areas within one brain hemisphere. These types of seizures usually result from a condition that is present at birth or in early infancy.
- **Functional hemispherectomy.** This procedure is mostly performed on children to remove the connecting nerves without removing actual pieces of the brain.

Gynaecological and Andrological procedure

Hysterectomy

What is a hysterectomy?

A hysterectomy is the surgical removal of your uterus and, most likely, your cervix. Depending on the reason for the surgery, a hysterectomy may involve removing surrounding organs and tissues, like your fallopian tubes and ovaries.

You won't be able to get pregnant and you won't get your menstrual period after a hysterectomy.

What are the different kinds of hysterectomy?

Your healthcare provider will discuss which type of hysterectomy you need, depending on your condition. This will determine if they'll also need to remove your fallopian tubes and/or ovaries.

The different types of hysterectomy surgeries are:

- **Total hysterectomy:** Removing your uterus and cervix but leaving your ovaries. Because you still have your ovaries, you won't enter menopause immediately after surgery.
- **Supracervical hysterectomy:** Removing just the upper part of your uterus while leaving your cervix. You may also have your fallopian tubes and ovaries removed at the same time. Because you still have your cervix, you'll still need Pap smears.
- **Total hysterectomy with bilateral salpingo-oophorectomy:** Removing your uterus, cervix, fallopian tubes (salpingectomy) and ovaries (oophorectomy). Removing your ovaries will start menopause immediately (if you haven't already completed it).
- **Radical hysterectomy with bilateral salpingo-oophorectomy:** The removal of your uterus, cervix, fallopian tubes, ovaries, the upper portion of your vagina, and some surrounding tissue and lymph nodes. Providers use this type most commonly when cancer is involved. You'll enter menopause after this type because you won't have ovaries anymore.

What is a laparotomy?

A laparotomy is a surgical procedure that opens up your abdomen to expose your organs.

Sometimes, a laparotomy is exploratory — a way for your surgeon to see what's going on inside your body and possibly take tissue samples in order to diagnose your condition. It can also be therapeutic — a way of treating some conditions. You may need open surgery to remove an organ or to manage a critical condition, or your surgeon may find and fix a problem while they're exploring.

The term "laparotomy" actually refers to the incision through your abdominal wall that opens up your peritoneal cavity — the cavity that includes your abdomen and pelvis. But you'll often hear the term used to describe an open exploratory procedure. Other terms for laparotomy include "celiotomy" (an incision into your abdomen) and "peritoneotomy" (an incision into your peritoneal cavity).

What is the difference between a laparotomy vs. a laparoscopy?

Laparotomy and laparoscopy are two different methods for surgically exploring the inside of your abdomen.

Laparoscopic surgery is an alternative to open surgery. Instead of opening up your abdomen to expose your organs, laparoscopic surgery works by inserting a laparoscope, a tiny lighted camera, through a small "keyhole" incision in your

abdomen. Your surgeon operates through other keyholes, using the camera as a guide. A laparoscopy is an exploratory surgery similar to a laparotomy, using a laparoscope.

Each method has its benefits and drawbacks. Laparoscopy is less invasive and easier to recover from, making smaller wounds and leaving smaller scars. However, it requires special training and equipment, and it takes longer, which may not work in an emergency. A laparotomy is a standard procedure that any hospital can provide. In certain cases, it's also more practical for managing critical conditions.

Why would you need a laparotomy?

Your surgeon might need to explore your peritoneal cavity to find the source of a problem, such as unexplained abdominal pain or internal bleeding, if they haven't been able to find it on imaging tests. They might need access to everything if they don't know where the issue is located or how widely it might be spread. They might choose laparotomy over laparoscopy if your condition appears urgent.

Laparotomy is often an emergency procedure, but not always. You might have a laparotomy for a planned or unplanned C-section. Your surgeon might schedule a laparotomy to remove an organ or to remove cancer. It's also helpful for cancer staging, to find out how far cancer may have spread from its original location. This is a "staging laparotomy." They also might need to take tissue samples for biopsy.

Respiratory Syndromes

Chronic obstructive pulmonary

Chronic obstructive pulmonary disease (COPD) is an ongoing lung condition caused by damage to the lungs. The damage results in swelling and irritation, also called inflammation, inside the airways that limit airflow into and out of the lungs. This limited airflow is known as obstruction. Symptoms include trouble breathing, a daily cough that brings up mucus and a tight, whistling sound in the lungs called wheezing.

COPD is most often caused by long-term exposure to irritating smoke, fumes, dust or chemicals. The most common cause is cigarette smoke.

Emphysema and chronic bronchitis are the two most common types of COPD. These two conditions usually occur together and can vary in severity among people with COPD.

Chronic bronchitis is inflammation of the lining of the tubes that bring air into the lungs. These tubes are called bronchi. The inflammation prevents good airflow into and out of the lungs and makes extra mucus. In emphysema, the small air sacs of the lungs, called alveoli, are damaged. The damaged alveoli can't pass enough oxygen into the bloodstream.

Although COPD is a condition that can get worse over time, COPD is treatable. With proper management, most people with COPD can control symptoms and improve their quality of life. Proper management also can lower the risk of other conditions linked to COPD, such as heart disease and lung cancer.

Symptoms

COPD symptoms often don't appear until a lot of lung damage has occurred. Symptoms usually worsen over time, especially if smoking or other irritating exposure continues.

Symptoms of COPD may include:

- Trouble catching your breath, especially during physical activities.
- Wheezing or whistling sounds when breathing.

- Ongoing cough that may bring up a lot of mucus. The mucus may be clear, white, yellow or greenish.
- Chest tightness or heaviness.
- Lack of energy or feeling very tired.
- Frequent lung infections.
- Losing weight without meaning to. This may happen as the condition worsens.
- Swelling in ankles, feet or legs.

People with COPD also are likely to have times when their symptoms become worse than the usual day-to-day variation. This time of worsening symptoms is called an exacerbation (eg-zas-er-bay-shun). Exacerbations can last for several days to weeks. They can be caused by triggers such as smells, cold air, air pollution, colds or infections. Symptoms may include:

- Working harder than usual to breathe or having trouble breathing.
- Chest tightness.
- Coughing more often.
- More mucus or changes in mucus color or thickness.
- Fever.

Tuberculosis

Tuberculosis, also called TB, is a serious illness that mainly affects the lungs.

The germs that cause tuberculosis are a type of bacteria.

Tuberculosis can spread when a person with the illness coughs, sneezes or sings. This can put tiny droplets with the germs into the air. Another person can then breathe in the droplets, and the germs enter the lungs.

Tuberculosis spreads easily where people gather in crowds or where people live in crowded conditions. People with HIV/AIDS and other people with weakened immune systems have a higher risk of catching tuberculosis than people with typical immune systems.

Medicines called antibiotics can treat tuberculosis. But some forms of the bacteria no longer respond well to treatments.

Symptoms

When TB germs survive and multiply in the lungs, it is called a TB infection. A TB infection may be in one of three stages. Symptoms are different in each stage.

Primary TB infection. The first stage is called the primary infection. Immune system cells find and capture the germs. The immune system may completely destroy the germs. But some captured germs may still survive and multiply. Most people don't have symptoms during a primary infection. Some people may get flu-like symptoms, such as:

- Low fever.
- Tiredness.
- Cough.

Latent TB infection. Primary infection is usually followed by the stage called latent TB infection. Immune system cells build a wall around lung tissue with TB germs. The germs can't do any more harm if the immune system keeps them under control. But the germs survive. There are no symptoms during latent TB infection.

Active TB disease. Active TB disease happens when the immune system can't control an infection. Germs cause disease throughout the lungs or other parts of the body. Active TB disease may happen right after primary infection. But it usually happens after months or years of latent TB infection.

Symptoms of active TB disease in the lungs usually begin gradually and worsen over a few weeks. They may include:

- Cough.
- Coughing up blood or mucus.
- Chest pain.
- Pain with breathing or coughing.
- Fever.
- Chills.
- Night sweats.
- Weight loss.
- Not wanting to eat.
- Tiredness.
- Not feeling well in general.